- Tajikistan is mineral-rich and has some uranium deposits.
- It has substantial issues with legacy waste from past uranium mining and milling as a regional centre.

Electricity sector

Total generation (in 2019): 20.7 TWh

Generation mix: hydro 19.2 TWh (93%); coal 1.5 TWh (7%).

Privacy - Terms

Import/export balance: 2.9 TWh net export (0.3 TWh imports; 3.2 TWh exports)

Total consumption: 15.1 TWh

Per capita consumption: c. 1600 kWh in 2019

Source: International Energy Agency and The World Bank. Data for year 2019.

Tajikistan is immediately south of Uzbekistan and Kyrgyzstan (which in turn lie south of Kazakhstan) with a complex border, especially around the northern Sughd oblast.

Most of Tajikistan's electricity is hydroelectric. Power shortages are reported, and about 40% of the country's electricity is used in aluminium smelting at Tursunzade.

In February 2017 a nuclear cooperation agreement was signed with Russia. It provides a legal basis for interaction between the two countries in the nuclear power sector for the first time in history, while outlining cooperation in: the design, construction, operation and decommissioning of research reactors; used fuel and radioactive waste management; rehabilitation of tailing storage areas and use of decommissioned uranium mining and reprocessing facilities (a major current issue – see below); production of radioisotopes; use of nuclear technology in industry, medicine and agriculture; and education and training.

Uranium mining and milling

During the Soviet era, uranium ore mined in Tajikistan, Uzbekistan, and Kyrgyzstan was milled into yellowcake at Tajikistan's Vostochnyy Rare Metal Industrial Association (Vostokredmet), previously known as the Leninabad Mining and Chemical Combine, in Chkalovsk, Sughd oblast. This was the first uranium processing plant in the Soviet Union.

Vostokredmet incorporated seven mines and five plants including plant B, and processed up to one million tonnes of uranium ore per year to produce yellowcake for the Soviet nuclear power industry and Soviet military. Reportedly, Chkalovsk once had the capability to convert uranium concentrate into uranium hexafluoride. The plant provided the material for the USSR's first nuclear weapon.

Chkalovsk is 10 km east from Khujand (Khodjend/Khodzhent and formerly Leninabad – the country's second largest city) on the Syr

Darya River. The Sughd oblast protrudes north into Uzbekistan and includes the densely-populated Fergana Valley with the Syr Darya River. It is much closer to Tashkent in Uzbekistan than to Dushanbe, the Tajik capital.

The Tajik uranium mines included Taboshar (40 km north of Khujand), Adrasman (20 km NE of Taboshar), Mailisui, Uigur and Tyuya-Muyun. Mining continued to 1992.

Uzbek uranium ore processed in the Leninabad/Vostokredmet uranium production centre came from several small volcanic vein deposits in the Fergana valley and Kazamazar uranium district from about 1946.

Uranium exploration

By law, the size of Tajikistan's uranium resources has been considered a state secret. Though the northern region of Sughd oblast around Chkalovsk may be depleted of uranium, Tajik officials claim that the Pamir region in the south and the sparsely-populated mountainous Gorno-Badakhshan autonomous region in the southeast (comprising nearly half the country) may contain substantial uranium resources. There was a Tajikistan report in the 2002 'Red Book', but nothing since then.

In 2008, the Tajik government amended legislation that had prohibited foreign companies from participating in the state's uranium industry, and Chinese, Indian and other foreign companies were discussing potential projects with the government. In 2014, it was reported that the Tajik government had endorsed a draft agreement between Areva Mines and the Tajik Main Geology Directorate to explore for uranium and rare earth metal deposits. It was also reported that the government had given China Guangdong Nuclear Power Group (now China General Nuclear Power Group, CGN) and Uranium Corporation of India Ltd. (UCIL) permission to explore for uranium deposits.

Legacy waste from uranium mining

Throughout the period of the Vostokredmet/Leninabad Combine's activities, 32 million cubic metres of tailings (33.7 Mt) and 2.7 million cubic metres (1.8 Mt) of below-ore-grade material was accumulated in Tajikistan. In 1990 these were stored in nine tailings dumps (a total area of 174 ha and with 250 TBq activity) and 21

dumps of untreated low-grade ores (a total area of 22 ha and with 7 TBq of activity) and rock waste which are spread over the extensive area of the combine's activities. ISL residues 450-600 m deep in two mining departments had activity of 4 TBq. Many of the tailings dams are considered an environmental risk.

Twelve dumps of the mining department No.2 accumulated over 1953-1980 are on the terraces of the Syr Darya river 0.3-6.5 km away from Khujand. These total 16 ha and have activity of 4 TBq.

Also close to Chkalovsk and the large-scale Leninabad plant a tailings dump in the town of Gafurov was operated in 1945-1950 in connection with a pilot plant for hydro-metallurgical processing of uranium ores. The 3.9 ha tailings dump is within the area of service and pilot production buildings. It was completely dried and in 1963 its surface was covered with a one-metre layer of inert soil. However, further measures to reduce radon release were being considered in 1991.

Then over 1949-1967 a 26.7 ha tailings dump some 2 km from the town of Gafurov was in operation. The volume of waste is 2.6 million cubic metres; its activity is 29 TBq. Half a metre of soil covers the surface. A dump for the tailings from the HMP-1 plant has been in operation since 1963 in the Digmai/Degmay cavity 4 km from the town of Gafurov. It occupies 69.2 ha and is filled with 19 million cubic metres of waste, the activity of which is 156 TBq.

The **Taboshar** uranium mine site is in the Fergana Valley on the southern edge of the Kuramin Mountains close the Uzbek-Tajik border about 40 km due north of Khujand. The site consists of a non-rehabilitated open mine pit, dismantled production buildings, and three tailing sites with 10 million tonnes of low-grade ore. The town of Taboshar with its 12,000 inhabitants is only four kilometres away. A dump of below-ore-grade material and rock waste with a volume of 1.2 million m^3 and an activity of 2.5 TBq remained on the site of the Taboshar mine after its 15-year operation (1950-1965). Four tailings dumps with a total volume of 4.5 million m^3 and a total area of 57.4 ha were formed at different times during the operation of a pilot hydrometallurgical plant there over 1949-1965. It treated ores from Meilisui, Uigur and Adrasman mines. The total activity of material in the tailings dumps is 36 TBq.

East in the Fergana Valley, eight dumps up to 10 km away from the settlement of Charkassar resulted from the operation of the **Adrasman** mine over 1954-89. Their total area is 3.4 ha and the 292,100 tonnes of waste has a total activity of 0.54 TBq. On the site of the mine only 50 metres from the settlement there is a 3.4 ha

tailings dump from operations over 1949-1958. The volume of waste in it is 102,000 m^3 with a total activity of 2.4 TBq.

Further east again, in the valley of the Maili-Suu River 15 km away from the town of **Mailisai** a tailings dump was in operation from 1947 to 1962, and contains 1.34 million cubic metres, with activity of 19 TBq. The surface of the tailings dump has half a metre of soil on it.

Rehabilitation of tailings

In June 2012, the Tajik parliament requested assistance from its neighbours and international organizations, beyond some funding already received from the European Economic Community (EEC). In July 2014 the government resolved to rehabilitate legacy tailings by 2024, and in that year it received some $6.5 million from the IAEA towards this.

Some tailings treatment has been undertaken with Russian involvement since 2009 based at Vostokredmet. In November 2016 Russia and Tajikistan signed a protocol on rehabilitation of areas affected by past uranium mining. The deputy minister of industry and new technologies, said "elimination" of the country's nuclear legacy is a matter of national importance, "because it concerns our children's health and environmental safety for generations to come." The general director of Russia's Federal Centre for Nuclear and Radiation Safety said that the protocol marked the "start of a new phase" in the countries' joint efforts to implement the remediation program.

In June 2015 the European Bank for Reconstruction and Development (EBRD) set up a fund to deal with radioactive contaminated material resulting from Soviet-era uranium mining and processing in the Kyrgyz Republic, Tajikistan and Uzbekistan. It said that the Environmental Remediation Account for Central Asia (ERA) "is being established at the request of the European Commission, which is providing an initial €8 million, with additional funding under consideration." The fund would be used to finance remediation of "high-priority sites." It added: "Many of the legacy sites are concentrated along the tributaries to the Syr Darya River, which runs through the densely populated Fergana Valley, the agricultural centre of the region, which is shared by the Kyrgyz Republic, Tajikistan and Uzbekistan."

In August 2016 the Kyrgyzstan government announced rehabilitation works to begin in mid-2017, in collaboration with

Tajikistan and Russia as well as the IAEA, and funded by the EBRD. The EBRD signed framework agreements with the Kyrgyz Republic and Tajikistan in January 2017. ARMZ's VNIPIPT has opened an office in the Kyrgyz capital, with a view to prioritizing work at Min-Kush and Kaji-Sai in Kyrgyzstan and a site near Taboshar in Tajikistan.

Early in 2017 a team from the IAEA's Coordination Group for Uranium Legacy Sites (CGULS) visited legacy sites, including exposed tailings bodies and remnants of uranium mills and other infrastructure, referred to as Map 1-9 (Chkalovsk), Degmay, and Taboshar. The visit was supported by the Nuclear and Radiation Safety Agency of the Academy of Sciences of Tajikistan and site operator State Enterprise Tajredmet.

In 2017 the Stategic Master Plan for Environmental Remediation of Uranium Legacy Sites in Central Asia, which focuses on Kyrgyzstan, Tajikistan and Uzbekistan, was adopted by the three countries, the IAEA, the European Commission, the EBRD and the Commonwealth of Independent States Economic Council. The plan, published in May 2018, aimed to accelerate remediation efforts and identified a need for an additional €130 million needed to fund the work, in addition to €30 million already raised. It identified seven former production sites as the highest priority, including Degmay and Istikol in Tajikistan. An updated plan was approved in September 2021 by the three countries, the EU, the EBRD, Rosatom, and the IAEA, which identified a funding gap of €40 million remained.

See also *Legacy tailings* section in the information page on Uranium in Kyrgyzstan.

Regulation

The Tajik Nuclear and Radiation Safety Agency is part of the Academy of Sciences, Dushanbe, and is assisted by the EU.

R&D

A 20kW Russian Argus research reactor was completed in 1991 in Dushanbe, but was never fuelled or run. In 2007 the government asked the IAEA to dismantle it.

Non-proliferation

Tajikistan is a party to the Non-Proliferation Treaty since 1994 and in 2004 ratified an Additional Protocol agreement with the IAEA. It has also ratified the Central Asia Nuclear Weapon Free Zone treaty, with Kyrgyzstan, Turkmenistan and Uzbekistan. In 2005 Tajikistan ratified the International Convention for the Suppression of Acts of Nuclear Terrorism.

Notes & references

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Related information

Kyrgyzstan
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